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ITAI 1378

September 24, 2026

Reflection Journal

In this lab, the CNN architecture uses three convolutional layers. The convolutional layers help the model learn different features and patterns from the Chihuahua and muffin images. After the convolutional layers, the features are passed into fully connected layers to classify the image as either a Chihuahua or a muffin. This is different from the traditional neural network we used in the previous workshop because that model mainly used fully connected layers to process the image. The CNN is designed more specifically for image classification because the convolutional layers can learn visual features from the images.

CNN performs better than the traditional neural network from the previous lab. In the previous lab, the traditional neural network had around 83% validation accuracy, while CNN had around 96% validation accuracy. This shows that CNN performed better. CNN can learn visual features and patterns from the images, so the model can learn and recognize the difference between a Chihuahua and a muffin more efficiently. In terms of training time, CNN can take more time to train because it has convolutional layers to process and learn features from the images. I did not record the exact training time for both models, so I cannot compare the exact difference in training time.

One of the challenges I had in this lab was understanding why the validation accuracy could change even when I reran the same baseline model with the same settings. The first time I ran the baseline model, it reached 96.67% validation accuracy, but when I reran the whole model again, it reached 100%. I learned that CNN training has some randomness, so the results may be slightly different each time the model is trained. Another challenge was understanding how changing different settings would affect CNN performance. In experiment 1, I increased the epochs from 10 to 20, the validation accuracy decreased to 90%. In experiment 2, I decreased the learning rate from 0.001 to 0.0001, and the validation accuracy decreased to 83.33%. In experiment 3, I removed the last convolutional layer, the model still reached 96.67% validation accuracy. Overall, I learned that changing the settings does not always improve the model, and even the same model can give slightly different results each time it is trained.

Model	Epoch	Learning rate	CNN layer	Change tested	Validation accuracy
Baseline	10	0.001	3	Original model	96.67%
Experiment 1	20	0.001	3	Increase epoch	90%
Experiment 2	10	0.0001	3	Lower lr	83.33%
Experiment 3	10	0.001	2	Removed third CNN layer	96.67%

The real-world application would be identifying plant disease in agriculture. Farmers could use a CNN model to classify images of crops and identify signs of plant diseases. The model could learn different visual features from images, such as changes in the color, shape, or appearance of leaves. A farmer could take a picture of a plant, and the model could classify whether the plant appears healthy or shows signs of a certain disease. This could help farmers identify problems earlier and decide which crops may need more attention.

One ethical consideration would be human oversight. Farmers should not rely on the CNN model 100% because the model could make mistakes when identifying a plant disease. A farmer or an agricultural expert should review the results before making an important decision. Another consideration is responsibility. If the model gives a wrong prediction and causes a farmer to lose crops or money, it could be difficult to determine who is responsible for the mistake. It could be the farmer who used the model, the developer who created it, or the company that provided it. This is why human oversight and clear responsibility are important when using AI models in agriculture.